

Question 1

Marks 15

- a) Find $\int x^2 \cos(x^3 - 1) dx$ 2
- b) Using integration by parts, or otherwise, evaluate $\int \cos^{-1} x dx$ 3
- c) Using the substitution $u^2 = e^x + 1$, or otherwise, find $\int \frac{e^{2x} dx}{\sqrt{e^x + 1}}$ 4
- d) Using partial fractions, or otherwise, find $\int \frac{dx}{4x^2 - 1}$ 3
- e) Evaluate $\int_3^4 \frac{x^2 + x - 4}{x - 2} dx$ 3

Question 2

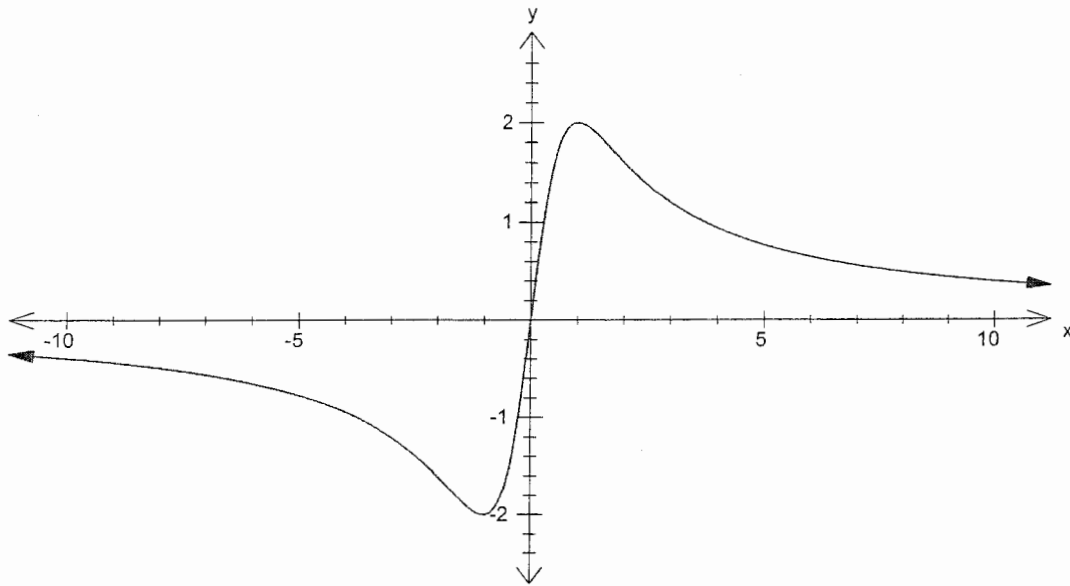
Marks 15

- a) Let $A = 2 - i$ and $B = 3 + 4i$.
Find, in the form $x + iy$
- (i) $A - iB$ 1
- (ii) $\bar{A}B$ 1
- (iii) $\frac{5}{A}$ 2
- b) If $z = \sqrt{3} + i$
- (i) Express z in modulus-argument form 2
- (ii) Hence find z^4 in $x + iy$ form. 2
- c) On an Argand diagram, clearly show the region where the inequalities
 $2 < |z| \leq 4$ and $-\frac{\pi}{4} \leq \arg z \leq \frac{\pi}{2}$ hold simultaneously. 3
- d) (i) With the aid of a diagram, describe the locus of Z on the Argand diagram if
 $\arg(z - 2k) - \arg(z) = \frac{\pi}{2}$, $k > 0$, $k \in R(\text{real numbers})$. 2
- (ii) What is the Cartesian equation of this locus? 2

Question 3

Marks 15

- a) The diagram below shows the graph of $y = f(x)$, which is an odd function.



Draw neat separate sketches showing all necessary detail of the following:

- | | |
|---|---|
| (i) $y = f(-x)$ | 1 |
| (ii) $y = [f(x)]^2$ | 1 |
| (iii) $y = \frac{1}{f(x)}$ | 2 |
| (iv) $y = x + f(x)$, showing any asymptotes. | 2 |
| (v) $y = f'(x)$ | 2 |
| b) Sketch the graph of $y = \frac{x-2}{x^2-4}$, clearly indicating any special features. | 3 |
| c) Consider the function $y = \tan^{-1}x - x + \frac{1}{3}x^3$. | |
| (i) Show that $\frac{dy}{dx} > 0$ for all values of $x > 0$. | 2 |
| (ii) Show that $\tan^{-1}x > x - \frac{1}{3}x^3$ for all values of $x > 0$. | 2 |

Question 4

Marks 15

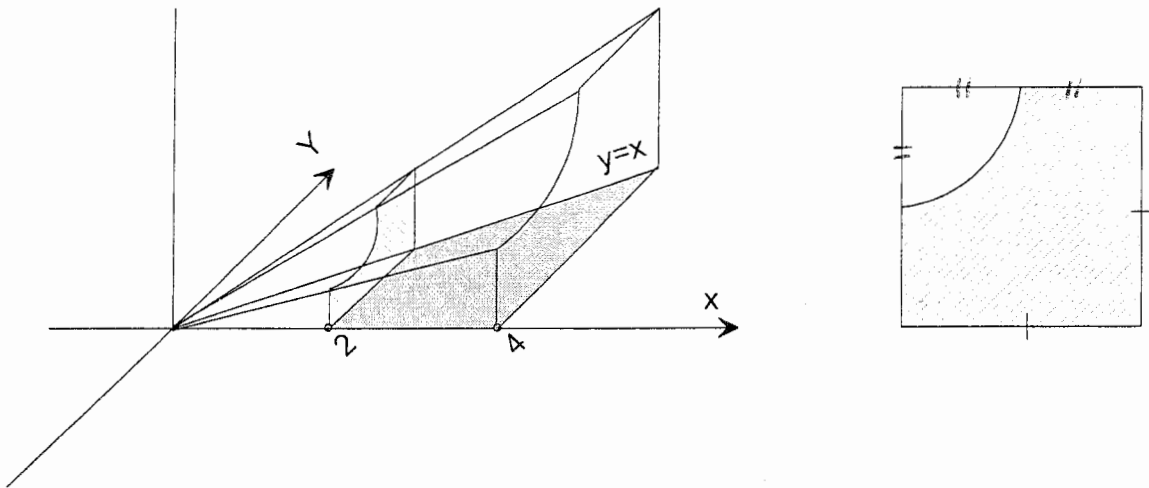
- a) An ellipse, E can be described as the locus of a point moving so that the sum of its distances from two fixed points (foci) is constant.
- (i) If the two fixed points are $S(4,0)$ and $S'(-4,0)$ and the sum of the distances of $P(x,y)$ from these points is 10 units, show that the equation of E is given by $\frac{x^2}{25} + \frac{y^2}{9} = 1$ [You may use the standard ellipse equation] 2
- (ii) Verify that $x = 5 \cos \theta$ and $y = 3 \sin \theta$ are the parametric equations of E . 1
- (iii) Find the equation of the normal to E at the point where $\theta = \frac{\pi}{6}$. 3
- (iv) Determine the eccentricity of E and, hence, the equations of the directrices. 2
- b) Given that α, β and γ are the roots of $3x^3 + 4x^2 - 2x - 1 = 0$, find the values of:
- (i) $\alpha + \beta + \gamma$ 1
- (ii) $\alpha\beta + \alpha\gamma + \beta\gamma$ 1
- (iii) $\alpha^2 + \beta^2 + \gamma^2$ 1
- (iv) $\alpha^3 + \beta^3 + \gamma^3$ 2
- c) Factorise $x^4 - 2x^2 - 15$ over the rational and complex fields. 2

Question 5

Marks 15

- a) The solid below has its base defined by the x -axis, the line $y = x$ and the lines $x = 2$ and $x = 4$ (metres). Cross-sections consist of a square with a quarter circle (quadrant) removed (as shown). The radius of the circle is half of the side length of the square. 3

Using the slicing technique, calculate the volume of this solid to the nearest cubic metre.



- b) (i) Show that, if $y = px + q$ is a tangent to the hyperbola 3

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1, \text{ then } p^2 a^2 - b^2 = q^2$$

- (ii) Hence or otherwise, find the equations of the tangents from the point $(1, 3)$ 3
to the hyperbola $\frac{x^2}{4} - \frac{y^2}{15} = 1$.

c) If $I_n = \int_0^1 x^n e^{-x} dx$

(i) Show that $I_n = -\frac{1}{e} + nI_{n-1}$ 3

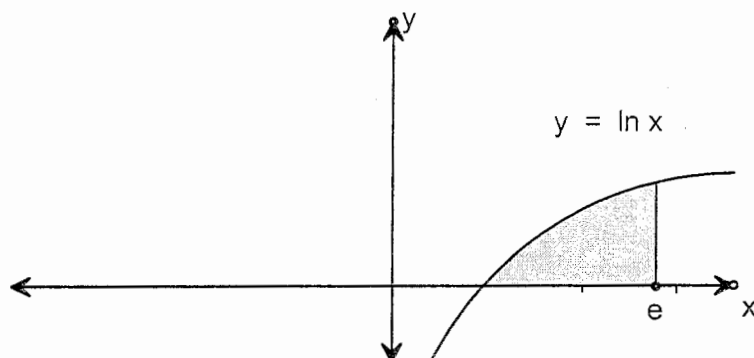
(ii) Hence find the exact value of $\int_0^1 x^3 e^{-x} dx$. 3

Question 6

Marks 15

- a) The region bounded by $y = \ln x$, $x = e$ and the x -axis is rotated about the y -axis. Use the cylindrical shells method to find the volume of the solid formed.

4



- b) The angles A , B and C are consecutive terms in an arithmetic sequence.

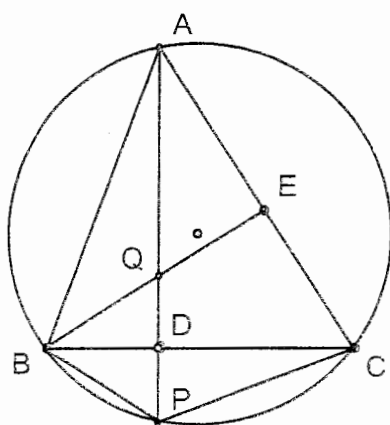
(i) Show that $A + C = 2B$

1

(ii) Hence, show that $\cos A \cos C - \cos^2 B = \sin A \sin C - \sin^2 B$.

2

- (iii) ABC is an acute angled triangle inscribed in a circle. AP is perpendicular to BC . Q is the point on AP such that $DQ = DP$. BQ produced meets AC at E .



(i) Copy the diagram showing the above information.

1

(ii) Show that $\triangle BDP \equiv \triangle BDQ$.

2

(iii) Show that $BDEA$ is a cyclic quadrilateral.

4

(iv) Show that BE is perpendicular to AC .

1

Question 7

Marks 15

- a) A particle is allowed to fall under gravity from rest in a medium which exerts a resistance proportional to the speed (v) of the particle.

- (i) Show that the particle reaches a terminal velocity, T given by 2

$$T = \frac{g}{k} \text{ (where } k \text{ is a positive constant).}$$

- (ii) Show that the distance fallen to reach half its terminal velocity ($\frac{T}{2}$) is given by 4

$$x = \frac{T^2}{g} \ln 2 - \frac{T^2}{2g}$$

- (iii) Determine an expression for the time taken to reach a speed of $\frac{T}{2}$. 3

- b) Consider the curve given by the equation $x^2 - y^2 + xy + 5 = 0$.

- (i) Show that $\frac{dy}{dx} = \frac{2x+y}{2y-x}$ 2

- (ii) Hence or otherwise, find the coordinates of the points on the curve where the tangent to the curve is parallel to the line $y = x$. 2

- c) By taking the logarithms of both sides of $y = U(x) \cdot V(x)$, verify the Product Rule for differentiation. 2

Question 8

Marks 15

- a) (i) If α is a double root of a polynomial $P(x)$, show that α is a zero of $P'(x)$. 2
- (ii) Find the integers m and n such that $(x + 1)^2$ is a factor of $x^5 + 2x^2 + mx + n$. 3
- b) (i) Find the five roots of $z^5 = 1$ and write them in mod-arg form. 3
- (ii) Show these roots on an Argand diagram and find the area (in exact form) of the pentagon formed by them. 2
- (iii) Factorise $z^5 - 1$ over the real field. 2
- c) The lengths of the sides of a triangle are the first three terms of an arithmetic sequence, with the first term equal to 1 and the common difference d . 3
- Find the set of possible values of d .

End of paper

SOLUTIONS — EXTENSION 2 TRIAL HSC — 2010

Q1/ a) $\frac{d}{dx} \sin(x^3-1) = 3x^2 \cos(x^3-1)$

$$\therefore \int x^2 \cos(x^3-1) dx = \frac{1}{3} \sin(x^3-1) + C$$

b) $\int \cos^{-1} x dx = x \cos^{-1} x - \int \frac{-x}{\sqrt{1-x^2}} dx$

$$= x \cos^{-1} x - \frac{1}{2} \int \frac{-2x}{\sqrt{1-x^2}} dx$$

$$= x \cos^{-1} x - \frac{1}{2} \int \frac{dw}{\sqrt{w}}$$

$$= x \cos^{-1} x - \frac{1}{2} \times 2w^{\frac{1}{2}} + C$$

$$= x \cos^{-1} x - \sqrt{1-x^2} + C$$

Let $u = x$
 $du = dx$
 $v = \cos^{-1} x$
 $dv = \frac{-1}{\sqrt{1-x^2}} dx$
 Let $w = 1-x^2$
 $dw = -2x dx$

c) $\int \frac{e^{2x} dx}{\sqrt{e^x+1}} = 2 \int \frac{e^x \cdot e^x dx}{2\sqrt{e^x+1}}$

$$= 2 \int e^x du$$

$$= 2 \int (u^2-1) du$$

$$= 2 \left(\frac{u^3}{3} - u \right) + C$$

$$= 2 \left[\frac{(e^x+1)^{\frac{3}{2}}}{3} - \sqrt{e^x+1} \right] + C$$

$$= \frac{2}{3} \left[(e^x+1)^{\frac{3}{2}} - 3\sqrt{e^x+1} \right] + C$$

$$= \frac{2}{3} \sqrt{e^x+1} ((e^x+1) - 3) + C = \frac{2}{3} \sqrt{e^x+1} (e^x-2) + C$$

$u^2 = e^x + 1$
 $u = (e^x+1)^{\frac{1}{2}}$
 $du = \frac{1}{2} (e^x+1)^{-\frac{1}{2}} \cdot e^x dx$
 $= \frac{e^x}{2\sqrt{e^x+1}} dx$

d) $\int \frac{dx}{4x^2-1} = \int \left(\frac{-\frac{1}{2}}{2x+1} + \frac{\frac{1}{2}}{2x-1} \right) dx$

$$= -\frac{1}{4} \ln(2x+1) + \frac{1}{4} \ln(2x-1) + C$$

$$= \frac{1}{4} \ln\left(\frac{2x-1}{2x+1}\right) + C$$

Let $x = \frac{1}{2}$
 $\therefore 2B = 1$
 $\therefore B = \frac{1}{2}$

Let $x = -\frac{1}{2}$
 $\therefore -2A = 1$
 $\therefore A = -\frac{1}{2}$

Let $\frac{A}{2x+1} + \frac{B}{2x-1} = \frac{1}{4x^2-1}$
 $\therefore A(2x-1) + B(2x+1) = 1$

$$e) \int_3^4 \frac{x^2 + x - 4}{x-2} dx$$

$$= \int_3^4 \left(x+3 + \frac{2}{x-2} \right) dx$$

$$= \left[\frac{x^2}{2} + 3x + 2 \ln(x-2) \right]_3^4$$

$$= \left[(8 + 12 + 2 \ln 2) - \left(\frac{9}{2} + 9 + 2 \ln 1 \right) \right]$$

$$= 20 + 2 \ln 2 - 13 \frac{1}{2}$$

$$= 6 \frac{1}{2} + 2 \ln 2$$

$$\begin{array}{r} x+3 \\ x-2 \overline{) x^2+x-4} \\ \underline{x^2-2x} \\ 3x-4 \\ \underline{3x-6} \\ 2 \end{array}$$

Q2 a) (i) $A - iB = 2 - i - i(3+4i)$
 $= 2 - i - 3i + 4$
 $= 6 - 4i$

(ii) $\bar{A}B = (2-i)(3+4i)$
 $= 6 + 8i + 3i - 4$
 $= 2 + 11i$

(iii) $\frac{5}{2-i} = \frac{5(2+i)}{(2-i)(2+i)}$
 $= \frac{5(2+i)}{4+1}$
 $= 2+i$

b) $z = \sqrt{3} + i$

(i) $|z| = \sqrt{3+1}$
 $= 2$

$\arg z = \tan^{-1} \frac{1}{\sqrt{3}}$
 $= \frac{\pi}{6}$

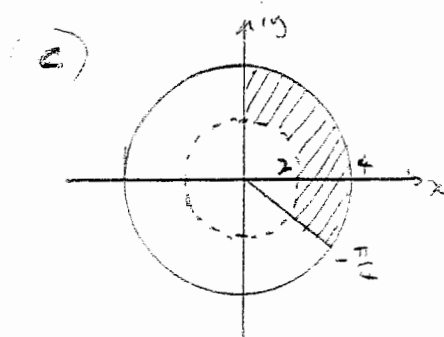
$\therefore z = 2 \cos \frac{\pi}{6}$

(ii) $z^4 = \left(2 \cos \frac{\pi}{6} \right)^4$

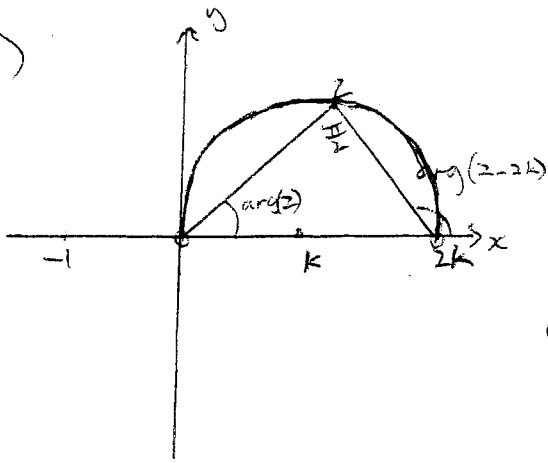
$$= 2^4 \cos \frac{4\pi}{6}$$

$$= 16 \cos \frac{4\pi}{6} + i 16 \sin \frac{4\pi}{6}$$

$$= -8 + 8\sqrt{3}i$$



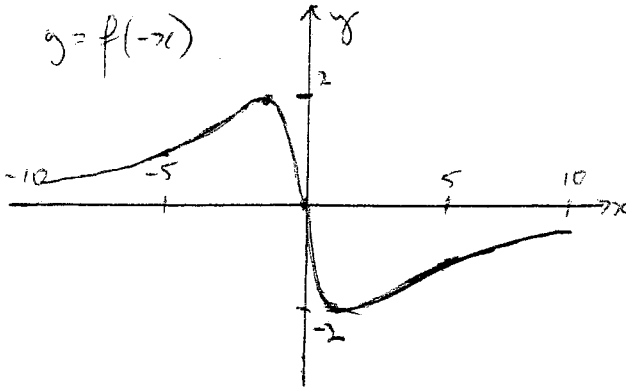
d) (i)



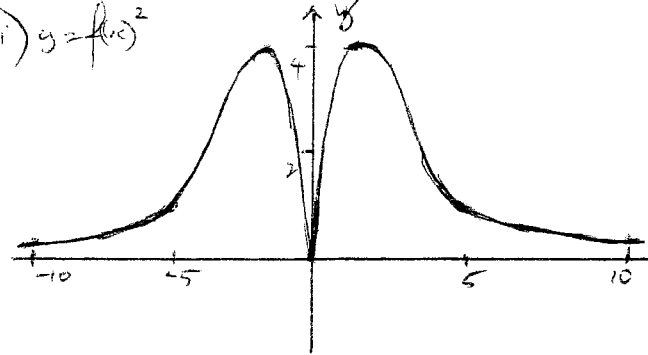
semi-
∴ locus of z is a circle centre $(k, 0)$
radius k .

(ii) Cartesian equation of locus
 $(x-k)^2 + y^2 = k^2$

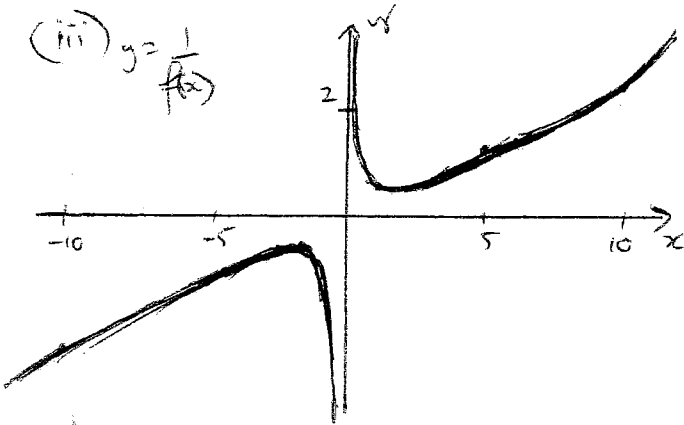
Q3 a) (i) $y = f(-x)$



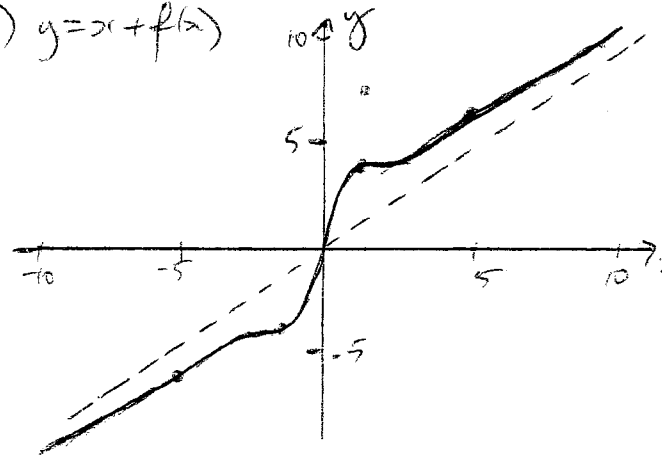
(ii) $y = f(x)^2$



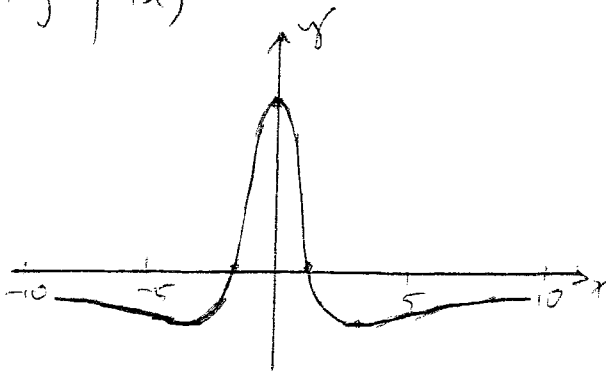
(iii) $y = \frac{1}{f(x)}$



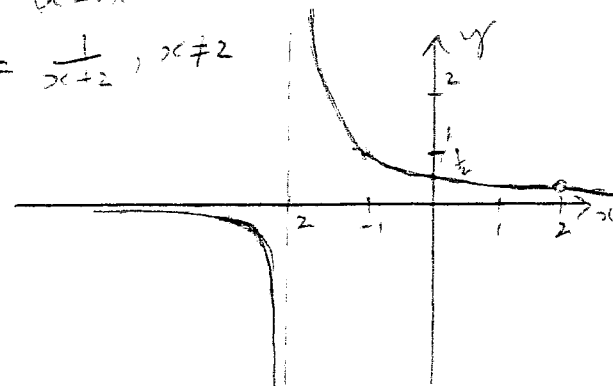
(iv) $y = x + f(x)$



(v) $y = f'(x)$



$$\begin{aligned} b) \quad y &= \frac{x-2}{x^2-4} \\ &= \frac{x-2}{(x-2)(x+2)} \\ &= \frac{1}{x+2}, \quad x \neq -2 \end{aligned}$$

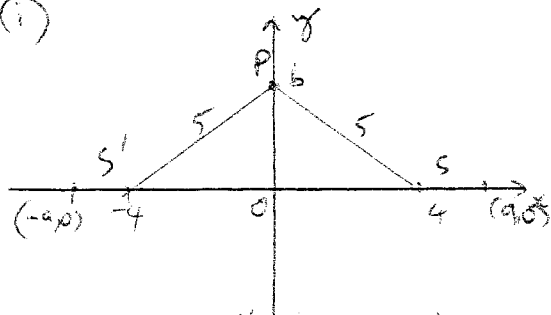


$$c) y = \tan^{-1} x - x + \frac{x^3}{3}$$

$$\begin{aligned} (i) \frac{dy}{dx} &= \frac{1}{1+x^2} - 1 + x^2 \\ &= \frac{1 + (x^2-1)(1+x^2)}{1+x^2} \\ &= \frac{1 + x^2 + x^4 - 1 - x^2}{1+x^2} \\ &= \frac{x^4}{1+x^2} \\ &> 0 \quad \forall x > 0. \end{aligned}$$

(ii) When $x=0$, $y=0$
and y is increasing for all $x > 0$
 $\therefore y > 0 \quad \forall x > 0$.
 $\therefore \tan^{-1} x - x + \frac{x^3}{3} > 0$ for $x > 0$
 $\therefore \tan^{-1} x > x - \frac{x^3}{3}$ for $x > 0$
Q.E.D.

Q4 a) (i)



P moves such that $PS + PS' = 10$
 $PO = 3$ (Pythag. Thm)
also $a = 5$ as $2a = 10$.
 $\therefore$ equation of ellipse is

$$\frac{x^2}{25} + \frac{y^2}{9} = 1$$

$$\begin{aligned} (iii) \frac{dx}{d\theta} &= -5 \sin \theta \\ \frac{dy}{d\theta} &= 3 \cos \theta \\ \frac{dy}{dx} &= \frac{dy}{d\theta} \cdot \frac{d\theta}{dx} \\ &= \frac{3 \cos \theta}{-5 \sin \theta} \end{aligned}$$

$$\begin{aligned} \text{When } \theta = \frac{\pi}{6}, \frac{dy}{dx} &= \frac{3 \times \frac{\sqrt{3}}{2}}{-5 \times \frac{1}{2}} \\ &= -\frac{3\sqrt{3}}{5} \end{aligned}$$

$$\therefore \text{grad normal} = \frac{5}{3\sqrt{3}}$$

$$\text{When } \theta = \frac{\pi}{6}, x = \frac{5\sqrt{3}}{2}, y = \frac{3}{2}$$

$$\begin{aligned} (ii) x &= 5 \cos \theta \quad y = 3 \sin \theta \\ \therefore \cos \theta &= \frac{x}{5} \quad \therefore \sin \theta = \frac{y}{3} \end{aligned}$$

$$\begin{aligned} \text{Now } \sin^2 \theta + \cos^2 \theta &= 1 \\ \therefore \left(\frac{x}{5}\right)^2 + \left(\frac{y}{3}\right)^2 &= 1 \quad (\text{reversed}) \\ \therefore \frac{x^2}{25} + \frac{y^2}{9} &= 1 \end{aligned}$$

$\therefore x = 5 \cos \theta$ and $y = 3 \sin \theta$
are the parametric equations for the ellipse.

$$\begin{aligned} \therefore \text{eqn of normal is} \\ y - \frac{3}{2} &= \frac{5}{3\sqrt{3}} \left(x - \frac{5\sqrt{3}}{2}\right) \\ \therefore 5x - 3\sqrt{3}y - 8\sqrt{3} &= 0 \end{aligned}$$

$$\begin{aligned} (iv) s &= ae \\ \therefore 4 &= 5e \\ \therefore e &= \frac{4}{5} \end{aligned}$$

$$\begin{aligned} \text{directrices} &\Rightarrow x = \pm \frac{a}{e} \\ &= \pm \frac{5}{\frac{4}{5}} \\ &= \pm \frac{25}{4} \end{aligned}$$

$$b) \quad 3x^3 + 4x^2 - 2x - 1 = 0$$

$$(i) \quad \sum \alpha = -\frac{b}{a} \quad \sum \alpha\beta = \frac{c}{a}$$

$$= -\frac{4}{3} \quad = -\frac{2}{3}$$

$$(iii) \quad \alpha^3 + \beta^3 + \gamma^3$$

As α, β and γ are roots,
we know...

$$3\alpha^3 + 4\alpha^2 - 2\alpha - 1 = 0$$

$$3\beta^3 + 4\beta^2 - 2\beta - 1 = 0$$

$$3\gamma^3 + 4\gamma^2 - 2\gamma - 1 = 0$$

$$\therefore 3(\alpha^3 + \beta^3 + \gamma^3) + 4(\alpha^2 + \beta^2 + \gamma^2) - 2(\alpha + \beta + \gamma) - 3 = 0$$

$$\therefore 3(\alpha^3 + \beta^3 + \gamma^3) + 4 \times \frac{28}{9} - 2 \times -\frac{4}{3} - 3 = 0$$

$$\therefore 3(\alpha^3 + \beta^3 + \gamma^3) = -\frac{109}{9}$$

$$\therefore \alpha^3 + \beta^3 + \gamma^3 = -\frac{109}{9}$$

$$c) \quad x^4 - 2x^2 - 15 = (x^2 - 5)(x^2 + 3) \quad \text{over } \mathbb{R}$$

$$= (x - \sqrt{5})(x + \sqrt{5})(x^2 + 3) \quad \text{over } \mathbb{R}$$

$$= (x - \sqrt{5})(x + \sqrt{5})(x - \sqrt{3}i)(x + \sqrt{3}i) \quad \text{over } \mathbb{C}$$

Q5

a) Area of cross-section, A

$$= \pi r^2 - \frac{1}{4} \pi \left(\frac{2r}{2}\right)^2$$

$$= \pi r^2 - \frac{\pi r^2}{16}$$

$$\therefore V = \int_2^4 x^2 \left(\frac{16-\pi}{16} \right) dx$$

$$= \frac{16-\pi}{16} \left[\frac{x^3}{3} \right]_2^4$$

$$= \frac{16-\pi}{16} \left[\frac{64}{3} - \frac{8}{3} \right]$$

$$= \frac{16-\pi}{16} \times \frac{56}{3}$$

$$= \frac{7(16-\pi)}{6} \text{ m}^3$$

$$\approx 15 \text{ m}^3$$

b) (i) Solve $y = px + q$ and $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ simultaneously.

$$\therefore \frac{x^2}{a^2} - \frac{(px+q)^2}{b^2} = 1$$

$$\therefore b^2 x^2 - a^2 (px+q)^2 = a^2 b^2$$

$$\therefore b^2 x^2 - a^2 (p^2 x^2 + 2pqx + q^2) = a^2 b^2$$

$$\therefore (b^2 - a^2 p^2) x^2 - 2apqx - a^2 q^2 - a^2 b^2 = 0$$

Because $y = px + q$ is tangential,
there is only one solution to this quadratic.

$$\therefore \Delta = 0$$

$$\therefore 4a^2 p^2 q^2 + 4(b^2 - a^2 p^2)(a^2 q^2 + a^2 b^2) = 0$$

$$\therefore a^4 p^2 q^2 + a^2 (b^2 - a^2 p^2)(q^2 + b^2) = 0$$

$$\therefore a^2 p^2 q^2 + b^2 q^2 + b^4 - a^2 p^2 q^2 - a^2 b^2 p^2 = 0$$

$$\therefore b^4 + b^2 q^2 - a^2 b^2 p^2 = 0$$

$$\therefore b^2 + q^2 - a^2 p^2 = 0$$

$$\therefore a^2 p^2 - b^2 = q^2 \quad \text{Q.E.D.}$$

(ii) $(1, 3)$ satisfies tangent

$$\therefore 3 = p + q$$

$$\therefore p = 3 - q$$

$$\text{Now } a^2 = 4 \text{ and } b^2 = 15$$

$$\therefore q^2 = 4p^2 - 15$$

$$\therefore q^2 = 4(3-q)^2 - 15$$

$$\therefore q^2 - 4(9 - 6q + q^2) + 15 = 0$$

$$\therefore -3q^2 + 24q - 21 = 0$$

$$\therefore 3q^2 - 24q + 21 = 0$$

$$\therefore 3(q-7)(q-1) = 0$$

$$\therefore q = 7 \text{ with } p = -4$$

$$\text{or } q = 1 \text{ with } p = 2$$

$$\therefore \text{tangents are } y = -4x + 7$$

$$\text{and } y = 2x + 1$$

c) (i) Let $u = x^n$ and $v = e^{-x}$

$$\therefore u' = nx^{n-1} \text{ and } v' = -e^{-x}$$

$$\therefore I_n = \left[-e^{-x} x^n \right]_0^1 + \int_0^1 nx^{n-1} e^{-x} dx$$

$$= \left[-\frac{1}{e} - 0 \right] + n I_{n-1}$$

$$= -\frac{1}{e} - n I_{n-1}$$

$$(ii) \int_0^1 x^3 e^{-x} dx = I_3$$

$$= -\frac{1}{e} + 3I_2$$

$$I_2 = -\frac{1}{e} + 2I_1$$

$$I_1 = -\frac{1}{e} + 1I_0$$

$$I_0 = \int_0^1 e^{-x} dx$$

$$= [e^{-x}]_0^1$$

$$= 1 - \frac{1}{e}$$

$$\therefore I_1 = 1 - \frac{2}{e}$$

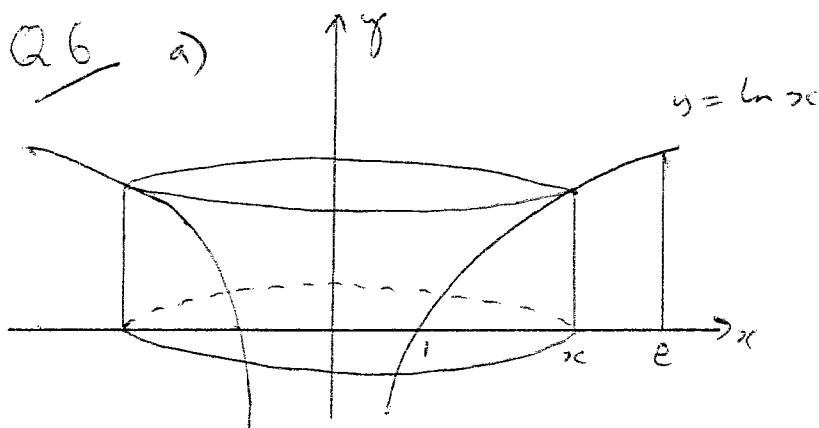
$$\therefore I_2 = 2\left(1 - \frac{2}{e}\right) - \frac{1}{e}$$

$$= 2 - \frac{5}{e}$$

$$\therefore I_3 = 3\left(2 - \frac{5}{e}\right) - \frac{1}{e}$$

$$= 6 - \frac{16}{e}$$

$$\therefore \int_0^1 x^3 e^{-x} dx = 6 - \frac{16}{e}$$



$$\text{let } u = \ln x$$

$$du = \frac{1}{x} dx$$

$$\text{let } v = \frac{x^2}{2}$$

$$\therefore dv = x dx$$

b) (i) $A, B, C \rightarrow AP$

$$A, A+d, A+2d$$

$$A+C = 2A+2d$$

$$2B = 2(A+d)$$

$$= 2A+2d$$

$$\therefore A+C = 2B$$

(ii)

$$\therefore \cos(A+C) = \cos 2B$$

$$\therefore \cos A \cos C - \sin A \sin C = \cos^2 B - \sin^2 B$$

$$\therefore \cos A \cos C - \cos^2 B = \sin A \sin C - \sin^2 B$$

$$A_{\text{shell}} = 2\pi x \ln x$$

$$\therefore V = \int_1^e 2\pi x \ln x dx$$

$$= 2\pi \int_1^e x \ln x dx$$

$$= 2\pi \left[\frac{x^2}{2} \ln x \right]_1^e + 2\pi \int_1^e \frac{x^2}{2} \cdot \frac{1}{x} dx$$

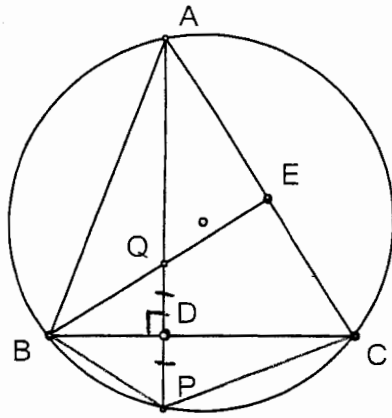
$$= 2\pi \left[\frac{e^2}{2} - 0 \right] - \pi \left[\frac{x^2}{2} \right]_1^e$$

$$= \pi e^2 - \pi \left[\frac{e^2}{2} - \frac{1}{2} \right]$$

$$= \pi \left[\frac{e^2}{2} + \frac{1}{2} \right]$$

$$= \frac{\pi}{2} [e^2 + 1] u^3$$

Q) (i)



(ii) In $\triangle BDQ$ and $\triangle BDP$,
 $DP = DQ$ (given)
 BD is common
 $\angle BDQ = \angle BDP = 90^\circ$ (given)
 $\therefore \triangle BDQ \equiv \triangle BDP$ (SAS)

(iii) $\angle BQD = \angle BPD$ (corresponding angles in congruent triangles)

$\therefore \angle EBQ = \angle PBC$ (same angles)

Now $\angle PAC = \angle PBC$ (two angles on same arc)

$\therefore \angle EBQ = \angle PAE$ ($\angle PAE$ same angle as $\angle PAC$)

$\therefore BDEA$ is a cyclic quadrilateral (two angles on same arc DE)

(iv) $\angle BDA = \angle BEA$ (two angles on same arc, AB)

$\angle BDA = 90^\circ$ (given)

$\therefore \angle BEA = 90^\circ$

$\therefore BE \perp AC$

Q7 a) (i) $\ddot{x} = g - kv$, $k > 0$
 terminal velocity, $\ddot{x} = 0$
 $\therefore g - kv = 0$
 $\therefore g = kv$
 $\therefore v = \frac{g}{k}$

$\therefore$ terminal velocity, $T = \frac{g}{k}$

When $v = \frac{T}{2}$,

$$x = \frac{T}{k} \cdot \ln 2 - \frac{T}{2k}$$

but $k = \frac{g}{T}$

$$\therefore x = \frac{T^2}{g} \cdot \ln 2 - \frac{T^2}{2g}$$

(ii) $\ddot{x} = g - kv$
 $\therefore \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = g - kv$

$$\therefore v \frac{dv}{dx} = g - kv$$

$$\therefore \frac{dv}{dx} = \frac{g - kv}{v}$$

$$\therefore \frac{dx}{dv} = \frac{v}{g - kv}$$

$$= \frac{1}{k} \left(\frac{v}{\frac{g}{k} - v} \right)$$

$$= \frac{1}{k} \left(\frac{v}{T - v} \right) \text{ as } T = \frac{g}{k}$$

$$= \frac{1}{k} \left(-1 + \frac{T}{T - v} \right)$$

$$\therefore x = \frac{1}{k} \left[-v - T \ln(T - v) \right] + c$$

When $x = 0$, $v = 0$

$$\therefore 0 = \frac{1}{k} \left[-T \ln T \right] + c$$

$$\therefore c = \frac{T \ln T}{k}$$

$$\therefore x = \frac{1}{k} \left[-v - T \ln(T - v) + T \ln T \right]$$

$$= \frac{T}{k} \cdot \ln \left(\frac{T}{T - v} \right) - \frac{v}{k}$$

$$(iii) \quad \dot{z} = \frac{dv}{dt} = g - kv$$

$$\therefore \frac{dt}{dv} = \frac{1}{g - kv}$$

$$\therefore t = -\frac{1}{k} \ln(g - kv) + C$$

$$\text{When } t=0, v=0$$

$$\therefore 0 = -\frac{1}{k} \ln g + C$$

$$\begin{aligned} \therefore t &= -\frac{1}{k} \ln(g - kv) + \frac{1}{k} \ln g \\ &= \frac{1}{k} \ln \left(\frac{g}{g - kv} \right) \end{aligned}$$

$$\text{When } v = \frac{T}{2}$$

$$= \frac{g}{2k} \quad \text{as } T = \frac{g}{k}$$

$$\therefore t = \frac{1}{k} \ln \left(\frac{g}{g - k \cdot \frac{g}{2k}} \right)$$

$$= \frac{1}{k} \ln \left(\frac{1}{1 - \frac{1}{2}} \right)$$

$$= \frac{1}{k} \ln 2$$

$$= \frac{T}{g} \ln 2 \quad \text{as } k = \frac{g}{T}$$

$$b)(i) \quad x^2 - y^2 + xy + 5 = 0$$

$$\therefore 2x - 2y \frac{dy}{dx} + y + x \frac{dy}{dx} = 0$$

$$2x + y - \frac{dy}{dx} (2y + x) = 0$$

$$\frac{dy}{dx} (2y + x) = 2x + y$$

$$\therefore \frac{dy}{dx} = \frac{2x + y}{2y + x}$$

$$(ii) \quad \text{Tangent parallel to } y = x.$$

$$\therefore \frac{dy}{dx} = 1$$

$$\therefore 2x + y = 2y + x$$

$$\therefore y = x$$

$$\text{Sub into } x^2 - y^2 + xy + 5 = 0$$

$$\therefore x^2 - x^2 + 3x^2 + 5 = 0$$

$$-5x^2 + 5 = 0$$

$$\therefore x^2 = 1$$

$$\therefore \text{contact points are } (1, 3) \text{ and } (-1, 3)$$

$$c) \quad y = u(x) v(x)$$

$$\begin{aligned} \therefore \ln y &= \ln [u(x) v(x)] \\ &= \ln [u(x)] + \ln [v(x)] \end{aligned}$$

$$\therefore \frac{1}{y} \frac{dy}{dx} = \frac{u'(x)}{u(x)} + \frac{v'(x)}{v(x)}$$

$$\therefore \frac{1}{u(x)v(x)} \frac{dy}{dx} = \frac{u'(x)}{u(x)} + \frac{v'(x)}{v(x)}$$

$$\therefore \frac{dy}{dx} = \frac{u'(x)}{u(x)} u(x)v(x) + \frac{v'(x)}{v(x)} u(x)v(x)$$

$$= u'(x) v(x) + v'(x) u(x)$$

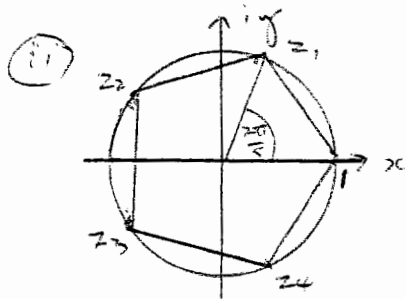
QED

Q8 a) (i) $P(x) = (x-2)^2 Q(x)$
 $\therefore P'(x) = (x-2)^2 Q'(x) + Q(x) \cdot 2(x-2)$
 $= (x-2) [(x-2) Q'(x) + 2Q(x)]$
 $\therefore 2$ is a root of $P'(x)$, as $x-2$ is a factor.

(ii) Let $P(x) = x^5 + 2x^2 + mx + n$
 $\therefore P'(x) = 5x^4 + 4x + m$
 Now $(x+1)^2$ is a factor of $P(x)$ and so $(x+1)$ is a factor of $P'(x)$
 $\therefore (-1)^5 + 2(-1)^2 + m(-1) + n = 0$ and $5(-1)^4 + 4(-1) + m = 0$
 $-1 + 2 - m + n = 0$ $5 - 4 + m = 0$
 $\therefore m - n = 1$ $\therefore m = -1$
 $\therefore n = -2$

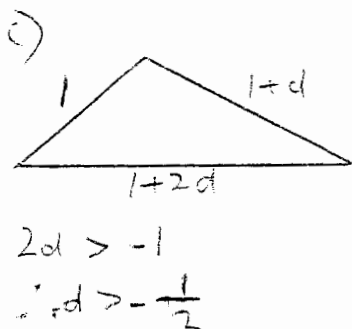
b(i) $z^5 = 1$
 Let $z^5 = (\cos \theta + i \sin \theta)^5$
 $= \cos 5\theta + i \sin 5\theta$
 $\therefore \cos 5\theta = 1$
 $\therefore 5\theta = 0, 2\pi, 4\pi, 6\pi, 8\pi$
 $\therefore \theta = 0, \frac{2\pi}{5}, \frac{4\pi}{5}, \frac{6\pi}{5}, \frac{8\pi}{5}$

$\therefore$ roots are $\cos 0 + i \sin 0 = 1$
 $\cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5}$
 $\cos \frac{4\pi}{5} + i \sin \frac{4\pi}{5}$
 $\cos \frac{6\pi}{5} + i \sin \frac{6\pi}{5} = \cos \frac{4\pi}{5} - i \sin \frac{4\pi}{5}$
 $\cos \frac{8\pi}{5} + i \sin \frac{8\pi}{5} = \cos \frac{2\pi}{5} - i \sin \frac{2\pi}{5}$



(iii) $z^5 - 1 = (z-1)(z - \cos \frac{2\pi}{5} - i \sin \frac{2\pi}{5})(z - \cos \frac{4\pi}{5} - i \sin \frac{4\pi}{5})(z - \cos \frac{6\pi}{5} - i \sin \frac{6\pi}{5})(z - \cos \frac{8\pi}{5} - i \sin \frac{8\pi}{5})$
 $= (z-1)(z^2 - 2z \cos \frac{2\pi}{5} + \sin^2 \frac{2\pi}{5} + \cos^2 \frac{2\pi}{5})(z^2 - 2z \cos \frac{4\pi}{5} + \sin^2 \frac{4\pi}{5} + \cos^2 \frac{4\pi}{5})$
 $= (z-1)(z^2 - 2z \cos \frac{2\pi}{5} + 1)(z^2 - 2z \cos \frac{4\pi}{5} + 1)$

$A = 5 \times \frac{1}{2} \times 1 \times 1 \times \sin \frac{2\pi}{5}$
 $= \frac{5}{2} \sin \frac{2\pi}{5} a^2$



Now, the sum of any 2 sides is greater than the third side.
 $\therefore 1 + (1+d) > 1+2d \quad | \quad 1 + (1+2d) > 1+d \quad | \quad (1+d) + (1+2d) > 1$
 $\therefore 2+d > 1+2d \quad | \quad \therefore 2+2d > 1+d \quad | \quad \therefore 2+3d > 1$
 $\therefore d < 1 \quad | \quad \therefore d > -1 \quad | \quad \therefore d > -\frac{1}{3}$

$\therefore -\frac{1}{3} < d < 1$ as all must be satisfied.